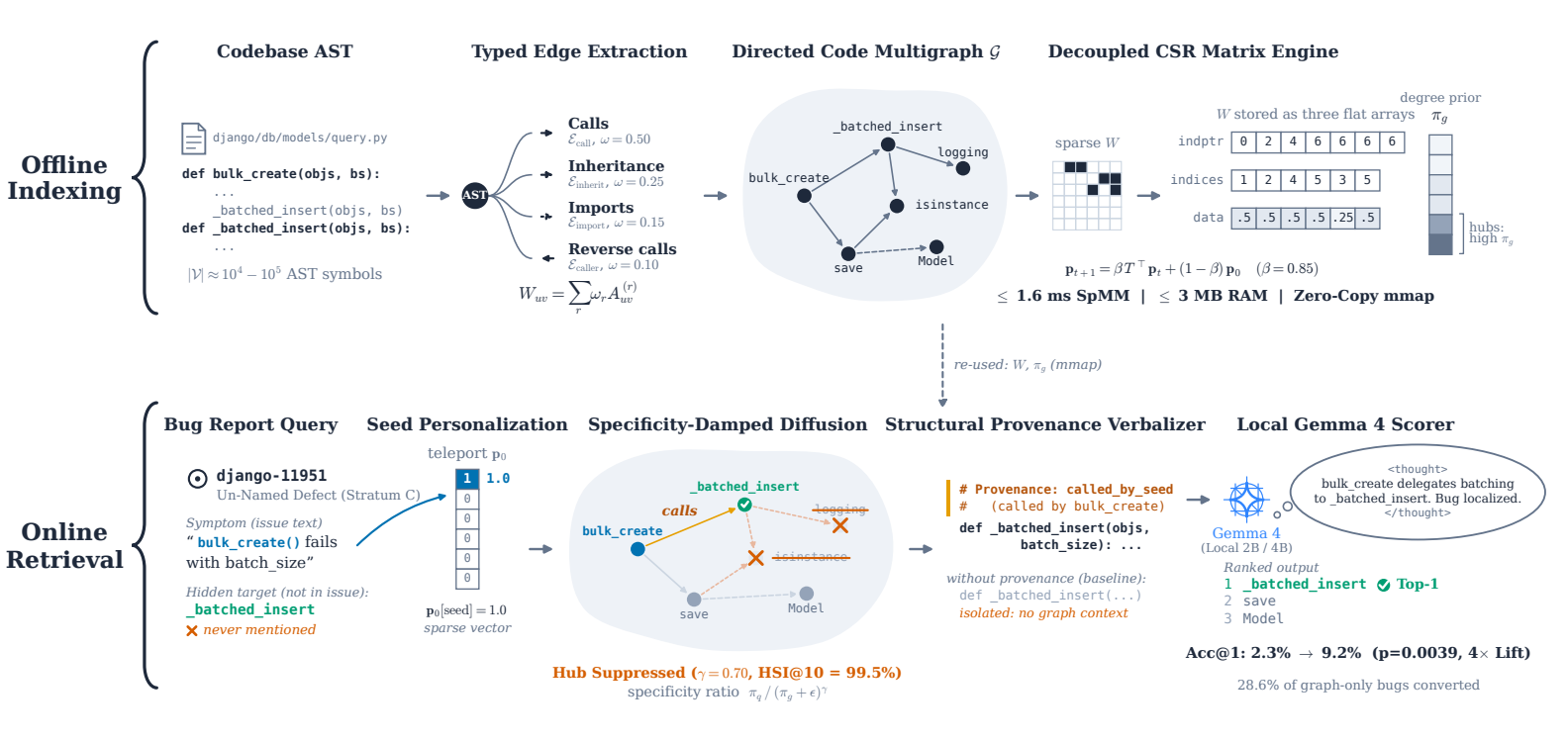

Offline Indexing
Codebase AST
django/db/models/query.py
def bulk_create(objs, bs):
...
_batched_insert(objs, bs)
def _batched_insert(objs, bs):
...
$|\mathcal{V}| \approx 10^4 - 10^5$ AST symbols
Typed Edge Extraction
AST
Calls
$\mathcal{E}_{\text{call}}$, $\omega = 0.50$
Inheritance
$\mathcal{E}_{\text{inherit}}$, $\omega = 0.25$
Imports
$\mathcal{E}_{\text{import}}$, $\omega = 0.15$
Reverse calls
$\mathcal{E}_{\text{caller}}$, $\omega = 0.10$
$W_{uv} = \sum_r \omega_r A_{uv}^{(r)}$
Directed Code Multigraph $\mathcal{G}$
_batched_insert
logging
bulk_create
isinstance
save
Model
Decoupled CSR Matrix Engine
sparse W
W stored as three flat arrays
indptr 0 2 4 6 6 6 6
indices 1 2 4 5 3 5
data .5 .5 .5 .5 .25 .5
degree prior π_g
hubs: high π_g
$\mathbf{p}_{t+1} = \beta T^\top \mathbf{p}_t + (1-\beta)\,\mathbf{p}_0 \quad (\beta = 0.85)$
≤ 1.6 ms SpMM | ≤ 3 MB RAM | Zero-Copy mmap
re-used: W, π_g (mmap)
Online Retrieval
Bug Report Query
django-11951
Un-Named Defect (Stratum C)
Symptom (issue text)
“bulk_create() fails with batch_size”
Hidden target (not in issue):
_batched_insert
never mentioned
Seed Personalization
teleport $\mathbf{p}_0$
1 1.0
0 0 0 0 0
$\mathbf{p}_0[\text{seed}] = 1.0$
sparse vector
Specificity-Damped Diffusion
_batched_insert
calls
logging
bulk_create
isinstance
save
Model
Hub Suppressed ($\gamma = 0.70$, HSI@10 = 99.5%)
specificity ratio $\pi_g / (\pi_g + \epsilon)^\gamma$
Structural Provenance Verbalizer
Provenance: called_by_seed
(called by bulk_create)
def _batched_insert(objs, batch_size): ...
without provenance (baseline):
def _batched_insert(...)
isolated: no graph context
Local Gemma 4 Scorer
<thought>
bulk_create delegates batching to _batched_insert. Bug localized.
</thought>
Gemma 4
(Local 2B / 4B)
Ranked output
1 _batched_insert Top-1
2 save
3 Model
Acc@1: 2.3% → 9.2% (p=0.0039, 4× Lift)
28.6% of graph-only bugs converted